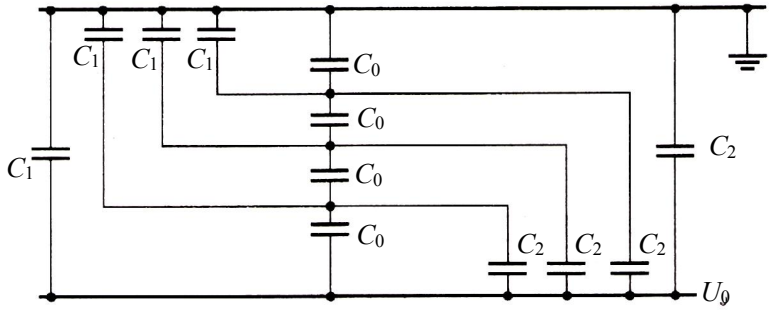
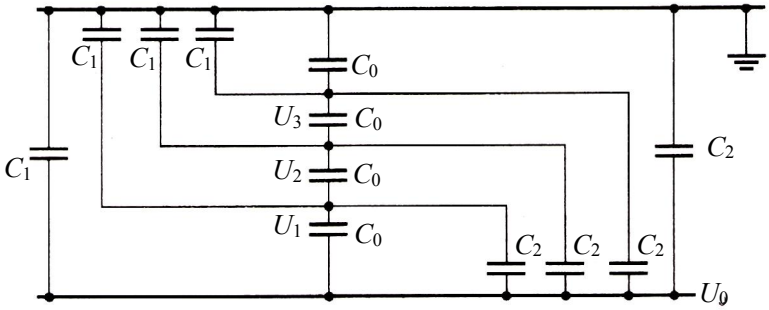


3. 중간 세 점의 전위를 각각 U_1 , U_2 , U_3라 하자 (그림). 등가 회로에서 각 점에 연결된 축전기 판 네 개의 전하량 대수합은 0이어야 하므로

$$\begin{cases}(U_1-U_2)C_0+U_1C_1-(U_0-U_1)C_0-(U_0-U_1)C_2=0,\\(U_2-U_3)C_0+U_2C_1-(U_1-U_2)C_0-(U_0-U_2)C_2=0,\\U_3C_0+U_3C_1-(U_2-U_3)C_0-(U_0-U_3)C_2=0.\end{cases} \quad (9)$$



네 절연체의 전압 합은 U_0 이어야 한다.

$$(U_0-U_1)+(U_1-U_2)+(U_2-U_3)+U_3=U_0 \ . \quad (10)$$

다음과 같이 놓으면

$$\triangle U_1=U_0-U_1 , \quad \triangle U_2=U_1-U_2 , \quad \triangle U_3=U_2-U_3 , \quad \triangle U_4=U_3 , \quad (11)$$

식 (9)를 정리하여

$$\begin{cases}\triangle U_1(C_0+C_1+C_2)-\triangle U_2C_0-U_0C_1=0,\\\triangle U_1(C_1+C_2)+\triangle U_2(C_0+C_1+C_2)-\triangle U_3C_0-U_0C_1=0,\\\triangle U_1(C_0+C_1+C_2)+\triangle U_2(C_0+C_1+C_2)+\triangle U_3(2C_0+C_1+C_2)-U_0(C_0+C_1)=0;\end{cases}$$

수치를 대입하면

$$\begin{cases}76\triangle U_1-70\triangle U_2-5U_0=0,\\6\triangle U_1+76\triangle U_2-70\triangle U_3-5U_0=0,\\76\triangle U_1+76\triangle U_2+146\triangle U_3-75U_0=0.\end{cases} \quad (12)$$

식 (12)를 풀면

$$\triangle U_1=0.298\ U_0 , \quad \triangle U_2=0.252\ U_0 , \quad \triangle U_3=0.228\ U_0 \ . \quad (13)$$